

Project Proposal

NFT Marketplace Website On Algorand Blockchain

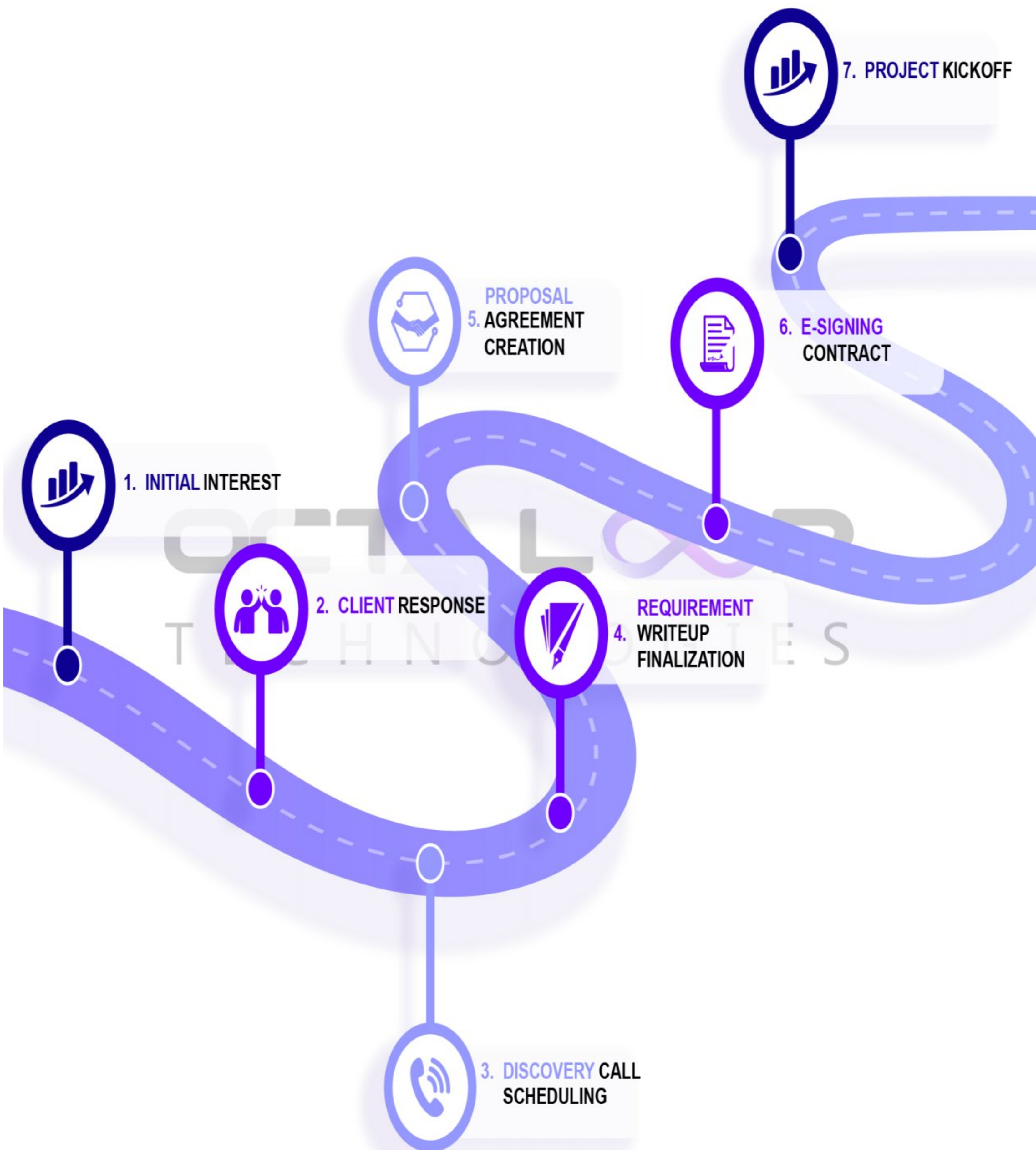
Client: Fry Foundation




Growth


Target


Global



Fry Foundation– NFT MARKETPLACE WEBSITE ON ALGORAND BLOCKCHAIN

We are pleased that you're considering **Octalooop Technologies** for development of “**NFT Marketplace Website On Algorand Blockchain**” please review the following outlines the details of this project proposal along with the pricing details.

This Agreement ("Agreement") is made and effective [18-July-2024], by and between [**Octalooop Technologies**] ("Development Team") and [**Fry Foundation**] ("Client").

DESCRIPTION OF SERVICES

Beginning on [18-July-2024], the Developer will provide the following services connected with the development of the “**NFT Marketplace Website On Algorand Blockchain**” (collectively, the “Services”):

Following things will be delivered upon completion of the Service:

- Fully functional Fry Foundation NFT Marketplace website on Algorand Blockchain
- Technical Documentation
- Complete Code Base

Our Portfolio and Relevant Experience

Octalooop Technologies (<https://octalooop.io>) is a versatile information technology firm that offers AI-based and web3 solutions to enterprises and businesses. Our dynamic team, comprising artists, designers, and developers, delivers a comprehensive range of IT services encompassing web and software development, cloud computing, and marketing strategies. We excel in AI and web3 design, prototyping, and development. A few of our portfolio projects are given below.

Projects:

- NFT Marketplace: <https://ametapark.vercel.app/>
- NFT Marketplace: <https://multiart.vercel.app/>
- Solana Trading Bot: <https://www.bonkbot.io/>
- Solana eBook NFT Marketplace: <https://www.soltype.io/>
- Solana Asset Management and Investment Platform: <https://solrise.finance/>
- Solana P2E Application: <https://beta.unlimitedcope.com/>
- Solana Trading Bot: <https://fluxbot.xyz/>
- Solana Based Decentralized Crypto Exchange: <https://www.crema.finance/>
- Solana Based NFT Project: <https://monkeykingdom.io/>
- Solana Based Decentralized Crypto Exchange: <https://www.step.finance/>
- Our Coming Soon Solana Based NFT Project: <https://degens.io/>
- Our P2E Game: <https://supervet.io/>
- Staking Dashboard: <https://moonrabbits-staking.vercel.app/>
- ICO dApp: <https://www.timevillage.io/token-sale>
- Our P2E Game: <https://gladiatorsverse.io/>



Client Requirements

Introduction

This section outlines the software requirements for the Fry Foundation NFT Marketplace, a platform for users to buy, sell, and create unique digital assets (NFTs) built on the Algorand blockchain. The platform will utilize the Fry Foundation's native token, FrysCrypto (FRY), for various functionalities.

1. Project Description

The Fry Foundation NFT Marketplace will offer the following core functionalities:

- **NFT Listing and Sales:** Users can list NFTs for fixed prices or create auctions.
- **FRY Integration:** FRY will be the primary token for listing fees, collection generation fees, and potentially future marketplace features.
- **AI NFT Generation:** Users can generate entire NFT collections based on text prompts through an integrated AI system.
- **User Interface (UI) and User Experience (UX):** The platform will prioritize a user-friendly and intuitive interface for both buying and selling NFTs, as well as generating collections.

2. Functional Requirements

2.1 User Management

- User registration and login with secure authentication mechanisms.
- User profiles displaying relevant information like collection ownership and transaction history.

2.2 NFT Management

- **Listing:** Users can create NFT listings specifying details like name, description, category, and traits.
- **Sales Options:**
 - Fixed price listings with a clear display of the FRY price.
 - Auction functionality with customizable starting price, duration, and reserve price (optional).
- **NFT Viewing:** Allow users to view detailed information about listed NFTs, including ownership history, metadata, and associated media.
- **Search and Filter:** Implement a search function with filters to navigate the NFT marketplace based on various criteria (e.g., category, collection, creator).

2.3 FRY Integration

- Secure integration with a supported Algorand wallet to facilitate FRY transactions.
- Users should be able to deposit and withdraw FRY for marketplace activities.
- Transparent display of FRY transaction fees associated with listing, collection generation, and potential future features.

2.4 AI NFT Generation

- Implement an AI system that generates unique NFTs based on user-provided text prompts.
- Users should be able to specify generation parameters like the number of NFTs in a collection and desired artistic style (if applicable).
- Collection generation requires a dedicated FRY fee.
- Clearly communicate the limitations and randomness inherent in AI-generated NFTs.

2.5 Security

- Secure storage of user credentials and private keys (using industry-standard practices).

- Smart contract audits to ensure the integrity and security of NFT ownership and transactions on the Algorand blockchain.

3. Non-Functional Requirements

3.1 Performance

- The platform should handle a high volume of users and transactions efficiently, minimizing latency and ensuring a smooth user experience.
- Fast loading times for NFT listings, searches, and user profiles.

3.2 Scalability

- The platform should be designed to accommodate future growth in user base and NFT listings. This includes consideration for potential upgrades to the underlying infrastructure.

3.3 Usability

- The user interface (UI) should be intuitive and easy to navigate for both novice and experienced users.
- Responsive design ensuring optimal user experience across various devices (desktop, mobile, tablet).

Proposed Solution

This section outlines a high-level approach to developing the NFT marketplace website on Algorand Blockchain, addressing the client's requirements feature-wise.

Development Phases

The development process can be divided into several phases:

- **Planning and Design**
 - Finalize technical specifications with the client, including detailed wireframes and mockups for the user interface (UI) and admin panel.
 - Define the smart contract architecture and functionalities for NFT management, pricing logic, royalty distribution, and marketplace operations.
- **Front-End Development**
 - Implementation of the UI mockups using React.js to create a dynamic and responsive user interface.
 - Integration with chosen Algorand wallet solution for user login and FRY transactions.
 - Develop components for NFT listings, search functionalities, user profiles, and collection generation prompts. Design intuitive interfaces for browsing NFTs, managing collections, creating listings, and purchasing NFTs.
- **Back-End Development**
 - Implement user authentication and authorization mechanisms using secure protocols.
 - Develop APIs for user interaction with the platform (listings, purchases, collection generation).
 - Integrate with the chosen Algorand wallet solution for FRY transactions and user balance management.
 - Connect with the chosen database to store user data, NFT metadata, transaction history, and collection generation parameters.
 - Implement user authentication and authorization mechanisms.

- **AI NFT Generation**

- Integrate AI models such as GPT-4o for generating NFTs.
- Develop a secure API integration between the backend and the chosen AI service.
- Design the collection generation functionality where users can provide text prompts and specify desired collection size and style (if supported by the AI service).
- Translate user prompts and parameters into a format compatible with the chosen AI service API.
- Process the generated NFT images and associated metadata received from the AI service.
- Store the generated NFTs securely within the platform and on the Algorand blockchain.

- **Smart Contract Development**

- Detailed analysis of business logic and functionalities for the marketplace, focusing on NFT ownership transfer, sales mechanisms (fixed price, auction), FRY integration, and potential future features.
- Smart contract design and development using Algorand Smart Contracts (ASCs) and Algorand SDKs. This includes writing code for:
 - Secure NFT minting process upon collection generation.
 - Ownership transfer logic for buying and selling NFTs.
 - Auction functionality (if applicable) for managing bids and determining winners.
- Integration with the chosen Algorand wallet solution for FRY fee collection and payout during transactions.

- **Deployment and Testing**

- Deploy the smart contracts to the Algorand testnet.
- Conduct comprehensive functional and security testing of the entire platform on the testnet.
- Refine and address any issues identified during testing.
- Deploy the platform to the Algorand mainnet after successful testing.

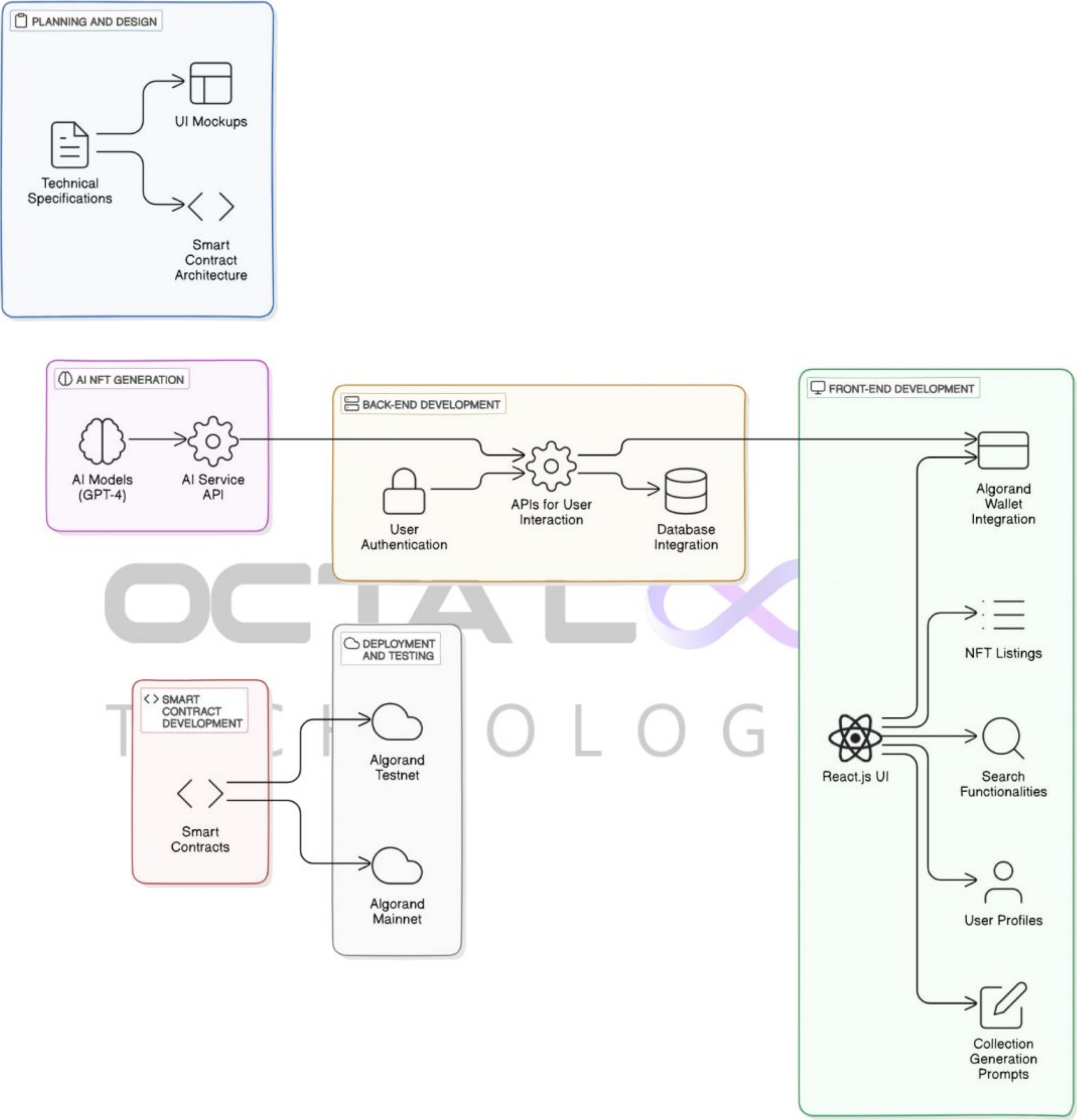
Technical Stack

- | | |
|---|---|
| <ul style="list-style-type: none">• Figma• React.js• Node.js• Web3.js• Python | <ul style="list-style-type: none">• MongoDB• IPFS• GPT-4o (more R&D)• AWS• Algorand SDK |
|---|---|



System Diagram

NFT Marketplace on Algorand Blockchain



Team Composition

Our company employs a hierarchical system to manage projects. For your project, we will assign a team comprising:

Sr#	Name	Designation	Experience
1	Noman Khan	Project Manager	4 Years
2	Qasim Amin	Technical Lead	5 Years
3	Taha Sharjeel	UI/UX Designer	5 Years
4	Tanzeel Hussain	Front-end Developer	4 Years
5	Shah Dost	Senior Blockchain Developer	5 Years
6	Qasim Amin	Associate Blockchain Developer	4 Years
7	Sheraz Alam	Full Stack Developer	5 Years
8	Harris Dilawar	Senior AI Developer	5 Years
9	Bilal Rafi	Q/A Engineer	4 Years

Project Management and Communication

In this project, we will implement the agile methodology. The Quality Assurance team will conduct parallel testing for each milestone to detect and address any bugs or issues. Task management will be handled through Trello/Jira, and communication within the company will take place via Slack.

The Project Manager will stay connected with you, delivering regular project updates through a WhatsApp channel or any other chat medium of your preference.

Maintenance and Support

Based on the call discussion with the client, Octaloop will provide three months of free support for bug fixes and minor changes in the already developed modules. However, if a new feature needs to be developed, separate charges will apply, added in a separate milestone.

After the initial three months, we will establish a new maintenance contract with the client on a monthly basis and provide daily support.

PAYMENT FOR SERVICES

In acknowledgment of the services to be executed by the development team, the Client undertakes to remunerate the development team for the services provided in the following manner:

- **Total Expense:** \$17,000
- **Initial Payment:** \$6,000
- **Outstanding Balance:** \$11,000 (Paid in installments based on progress)
- **Project Duration:** 45 Days

Note:

1. Progress payments will be implemented after the completion of each task, and the client is required to make payments following the amounts specified after the completion of each module.
2. Clients can make payments through the following methods:
 - Wise Payments
 - Payoneer Payments
 - Stripe Payments
 - Crypto Payments



Note: Commencement of work requires an upfront payment.

The fees for the Developer's services outlined in the Description of Services above will be invoiced based on the following schedule/breakdown:

INVOICE

Task#	Description	Timeline	Price (\$)
1	R&D, Technical Documentation, and UI/UX Designs in Figma	9 Days	2,000
2	Frontend Development of NFT Marketplace Website	9 Days	2,000
3	Backend Development (NFT collections, Gas fees handling, Homepage Development)	6 Days	3,000
4	Smart Contract Development (NFT Creation, Listing & Selling, Optional Royalty Payments)	8 Days	4,000
5	Backend Development (User Roles, Admin Dashboard Functionalities and Wallet Integration)	6 Days	3,000
6	Backend Development (AI text-to-image generation service integrated for NFT collection creation)	5 Days	2,000
7	Q/A, Testing & Refinement & Deployment	2 Days	1,000
Total:		45 Days	\$17,000

THANK YOU FOR YOUR BUSINESS



<https://www.octaloop.io/>



business@octaloop.io



Terms & Conditions

1. Mutual NDA:

Both parties agree to maintain the confidentiality of any proprietary information shared during the course of this project. This includes, but is not limited to, business plans, trade secrets, and project details.

2. Duties and Responsibilities:

The Client agrees to provide clear and detailed requirements for the software development project. The Developer agrees to utilize their best efforts and skills to deliver a software product that meets the specified requirements.

3. Training, Delivery, and Acceptance:

The Developer shall provide the Client with 10 Hours of necessary training to the Client's designated personnel upon completion of the software. The Client will thoroughly test the software within the free support period after delivery and developer will solve bugs for free in that support period.

4. Changes and Revisions:

Reasonable revisions to the software will be made at no additional cost if they fall within the initially agreed-upon scope. Any changes outside the scope will be subject to additional charges and a revised timeline.

5. Payment:

The Client commits to adhering to the payment schedule specified in the attached project proposal. An initial payment is required upfront, followed by subsequent progress payments upon the completion of project phases. Payment methods accepted include Wise, Stripe, Payoneer or Crypto Payments.

No-Refund Policy: Given the nature of progress payments, payments made for specific tasks are non-refundable once the corresponding task has been completed. Additionally, the Client will reimburse the Developer for all reasonable expenses on APIs/Libraries incurred during the Software development process.

6. Ownership of Software:

Upon full and final payment, the Client will own the rights to the software developed by the Developer. The Developer retains the right to use the software for portfolio and marketing purposes.

7. Code Delivery:

The Developer will provide the Client with the complete source code upon the finalization of the project and receipt of the final payment. However, code for each milestone will be provided upon client request once payment for that milestone is released.

8. Termination

Either party may terminate this Agreement with written notice if the other party breaches a material term and fails to remedy the breach within three days. Upon termination, the Client will pay for the services provided up to the termination date and Developer will deliver the software developed up to the termination date.



By placing their signatures below, both parties confirm their comprehension and acceptance of the terms mentioned in this agreement.

Client

Fry Foundation

**NFT Marketplace Website On Algorand
Blockchain**



Development Team

Aamir Ahmed (CEO)

Octaloop Technologies



OCTA LOOP
TECHNOLOGIES

